

Mahmoud Mamdoh Soliman

Data Engineer

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🌐 [Portfolio](#) 🔗 [LinkedIn](#) 🐙 [GitHub](#)

Professional Summary

Data Engineer with a strong foundation in architecting, optimizing, and automating data pipelines to drive business insights and operational efficiency. Skilled in building scalable data solutions using modern tools like Apache Airflow, dbt, and Power BI, with hands-on experience in ETL/ELT, data warehousing, and big data technologies. Passionate about delivering actionable insights and collaborating across teams to solve complex data challenges.

Skills

- **Programming Languages:** Python, SQL, Bash
- **Data Engineering & Big Data Tools:** Spark, Kafka, Airflow, Hadoop, Hive, dbt, Snowflake
- **Architecture & Concepts:** Medallion Architecture, Data Lakehouse, Data Warehouse, ETL/ELT Pipelines, Data Modeling, Streaming vs Batch Processing, Data Quality & Validation, Orchestration, and Data Governance
- **Data Transformation & Analytics:** PySpark, Pandas, SQLAlchemy, dbt Models, DAX, Power Query
- **Orchestration & Workflow Management:** Apache Airflow, Mage, Airbyte
- **Operating Systems:** Linux, Windows
- **Version Control & CI/CD:** Git, GitHub
- **Data Simulation & Streaming:** Kafka, Spark Streaming, Nifi
- **Tools & Utilities:** Power BI, SSIS, SSAS, Excel, Power Query

Professional Experience

- Part-Time Coding Instructor** | *iSchool, Egypt* Apr 2026 – Present
- Deliver live online coding sessions in Python for students aged 6–18
 - Simplify complex programming concepts into interactive, age-appropriate lessons to enhance student understanding
 - Guide students in building hands-on projects (e.g., simple games and applications) to reinforce practical skills
- Business Intelligence Analyst (Internship)** | *Addiction Treatments center Cairo, Egypt* Jul 2024 – Oct 2024
- Developed interactive Power BI dashboards that improved visibility into KPIs including customer satisfaction metrics, treatment success rates, and clinic operational efficiency, reducing manual reporting time by 30

Bootcamps & Specialized Training

- Data Engineering Zoomcamp** Jan 2025 – May 2025
DataTalks Club
- Hands-on program covering data ingestion, orchestration (Airflow), transformation (dbt), containerization (Docker)
 - Built a real-world ELT project using the Medallion Architecture and modern data stack tools
- Business Intelligence Track – Summer Training** Jul 2023 – Aug 2023
Information Technology Institute (ITI) | Menofia, Egypt
- Focused on building data warehouse solutions and business analytics using the Microsoft stack and open-source tools

Projects

Telecom Data Lakehouse Platform [GitHub](#)

Kafka, NiFi, MinIO, Iceberg (Hadoop Catalog), Spark, Airflow, ClickHouse, Python

- Designed and implemented an end-to-end event-driven data lakehouse for telecom use cases including calls, SMS, payments, recharges, and support tickets
- Streamed real-time events using Apache Kafka and ingested data into MinIO through Apache NiFi
- Ingested real-time events with Kafka and NiFi, and processed data using Spark with Apache Iceberg
- Developed Spark-based ETL pipelines for data validation, enrichment, and quality checks across Silver and Gold layers
- Designed a dimensional galaxy schema and loaded analytics-ready data into ClickHouse

Multisystem ETL Lakehouse for Business Insights [GitHub](#)

Airbyte, MinIO, Iceberg, Spark, Airflow, Python

- Engineered an end-to-end Data Lakehouse integrating Magento, Odoo, and Freshdesk using the Medallion Architecture
- Ingested and unified data with Airbyte, stored it in Iceberg on MinIO, and processed transformations with Spark
- Orchestrated ETL pipelines using Apache Airflow and monitored data health through a Streamlit dashboard

Telecom Data Analytics Hub (End-to-End Streaming Analytics Platform) [GitHub](#)

Kafka, Apache NiFi, PostgreSQL, dbt, Docker, SQL, Python, Power BI

- Designed and implemented an end-to-end telecom analytics platform using a medallion (Bronze/Gold) architecture
- Built real-time ingestion pipelines with NiFi and Kafka processing 5+ topics with validation, routing, and error handling
- Modeled analytics-ready galaxy schema with fact and dimension tables using dbt, including SCD Type 2 snapshots
- Enforced data quality using dbt tests (not_null, unique, relationships) with centralized failure storage
- Enabled business analytics by serving Gold-layer models to Power BI dashboards

Education

Bachelor of Computer Science and Mathematics | *Menofia University, Faculty of Science*

Sep 2021 – 2025

- Graduated 2025 | Relevant coursework: Data Structures, Algorithms, Database Systems, Statistics

Certifications & Professional Development

- **Data Engineering Zoomcamp** – DataTalks.Club (Jan-May 2025)
- **Business Intelligence Track** – Information Technology Institute (ITI), Menofia (Jul-Aug 2023)
- **Data Engineer Career Track** – DataCamp (Oct 2023)
- **Google Business Intelligence Professional Certificate** – Coursera (May 2023)
- **Data Engineering Foundations Specialization** – IBM (Nov 2023)